

SAT and Gröbner Bases

What the Algebraic Method Does and Does Not Buy

Version 2

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Abstract

Version 1 of this note proposed deciding satisfiability by arithmetizing a propositional formula φ as a polynomial $\tau(\varphi)$ over $\mathbf{F}_2$, forming the ideal $J = \langle \tau(\varphi) + 1, x_i^2 + x_i \rangle$, computing a Gröbner basis G for J , and testing whether 1 reduces to 0 modulo G . It suggested that the Gröbner basis step might be performed in polynomial time by the FGLM algorithm, and hence that $P = NP$.

The arithmetization and the ideal-membership test are correct, and are proved correct here. The complexity claim is not, and this version withdraws it. Two independent errors are identified: FGLM converts a Gröbner basis from one monomial order to another and cannot produce the first basis, which is the expensive object; and its running time is polynomial in the dimension of the quotient algebra $\mathbf{F}_2[x]/J$, which counts satisfying assignments and is exponential in the number of variables. Beyond these, the approach is closed by a theorem rather than left open: Gröbner basis refutation of J is exactly proof search in the Polynomial Calculus proof system, for which exponential lower bounds are known unconditionally.

What survives is worth stating. Polynomial Calculus polynomially simulates resolution, so nothing is lost by working algebraically; it refutes Tseitin formulas in polynomial size where resolution requires exponential size, so something is strictly gained on parity-structured instances; and degree- d refutations are findable in time $n^{O(d)}$, so the obstruction is degree rather than the algebraic setting. The correct research question is not whether this method decides SAT in polynomial time — it does not — but which families of formulas admit low-degree algebraic refutations.

Keywords: satisfiability, Gröbner basis, Boolean ideal, arithmetization, Polynomial Calculus, Nullstellensatz, proof complexity, resolution, Tseitin formulas, FGLM, degree lower bounds.

1 The construction, and what is right about it

Let φ be a propositional formula over $p_1, \dots, p_n$.

Definition 1.1 (Arithmetization). Define $\tau(\varphi) \in \mathbf{F}_2[x_1, \dots, x_n]$ by recursion:

$$\tau(p_i) := x_i, \quad \tau(\neg\varphi_1) := 1 + \tau(\varphi_1),$$

$$\tau(\varphi_1 \wedge \varphi_2) := \tau(\varphi_1)\tau(\varphi_2), \quad \tau(\varphi_1 \vee \varphi_2) := 1 + (1 + \tau(\varphi_1))(1 + \tau(\varphi_2)),$$

all arithmetic in $\mathbf{F}_2$. Put $J(\varphi) := \langle \tau(\varphi) + 1, x_1^2 + x_1, \dots, x_n^2 + x_n \rangle$.

Proposition 1.2 (Correctness). *For $a \in \{0, 1\}^n$ we have $\tau(\varphi)(a) = 1$ if and only if $a \models \varphi$. Consequently*

$$\varphi \text{ is unsatisfiable} \iff 1 \in J(\varphi) \iff \text{the reduced Gröbner basis of } J(\varphi) \text{ is } \{1\}.$$

Proof. Induction on φ . A variable is its own indicator. For negation, $1 + t = 1$ iff $t = 0$. For conjunction, a product of elements of $\{0, 1\}$ is 1 iff both are 1. For disjunction, $1 + (1 + t_1)(1 + t_2) = 1$ iff $(1 + t_1)(1 + t_2) = 0$ iff some $t_i = 1$.

The field equations $x_i^2 + x_i$ force every common zero into $\{0, 1\}^n$, so the variety of $J(\varphi)$ is the set of satisfying assignments. That variety is empty iff $J(\varphi)$ is the unit ideal — the ideal is zero-dimensional and radical, since it contains the field equations, so the weak Nullstellensatz applies without a radical correction — and an ideal is the unit ideal iff its reduced Gröbner basis in any monomial order is $\{1\}$. $\square$

Algorithm 1.3 (The procedure of Version 1). On input φ : compute $\tau(\varphi)$; form $J(\varphi)$; compute a Gröbner basis G of $J(\varphi)$; reduce 1 modulo G ; report *unsatisfiable* if the reduction is 0 and *satisfiable* otherwise.

Remark 1.4. Algorithm 1.3 is a correct decision procedure, by Proposition 1.2. Everything at issue is its running time, and that is entirely concentrated in the third step.

2 Why the complexity claim fails

Version 1 proposed to perform the Gröbner basis computation “in polynomial time using the FGLM algorithm.” There are two separate problems with this, and then a third that is fatal independently of the first two.

2.1 fglm is a change of order, not a computation from scratch

The algorithm of Faugère, Gianni, Lazard and Mora [4] takes as *input* a Gröbner basis of a zero-dimensional ideal with respect to one monomial order and returns a Gröbner basis with respect to another — in practice, from a fast degree order to the lexicographic order convenient for elimination. It presupposes the object whose cost is in question. Using it as the step that computes G is circular: the first basis must be obtained by Buchberger’s algorithm or a descendant such as F_4 or F_5 , and that is where the work is.

2.2 Its running time is polynomial in the wrong parameter

FGLM runs in time $O(nD^3)$, where

$$D = \dim_{\mathbf{F}_2}(\mathbf{F}_2[x_1, \dots, x_n]/J).$$

For a zero-dimensional radical ideal D is the number of points of the variety, so here D is the number of satisfying assignments of φ . This is at most 2^n and is routinely close to it. Being polynomial in D is not being polynomial in n : already the field equations alone give $\dim_{\mathbf{F}_2} \mathbf{F}_2[x]/\langle x_i^2 + x_i \rangle = 2^n$.

The mistake is a substitution of parameters. “Polynomial time” in the computer-algebra literature is frequently relative to the quotient dimension or to the Bézout number, not to the bit length of the input, and the two coincide only for ideals with few solutions.

2.3 The approach is closed by a theorem

The decisive point is not that the two objections above might be circumvented by a better algorithm. It is that the method as a whole is a known proof system with known lower bounds.

Definition 2.1 (Polynomial Calculus). Over a field K , the *Polynomial Calculus* F_{PC} has as its formulas the multilinear polynomials of $K[x_1, \dots, x_n]/\langle x_i^2 - x_i \rangle$ and as its rules

$$(f, g) \mapsto \alpha f + \beta g \quad (\alpha, \beta \in K), \quad f \mapsto x_i f.$$

A *refutation* of a set of polynomials is a derivation of 1. Its *size* is the total number of monomials and its *degree* is the largest total degree occurring in it.

Proposition 2.2. *Every run of Buchberger’s algorithm on $J(\varphi)$ that outputs $\{1\}$ is a F_{PC} refutation of the generators of $J(\varphi)$, of size at most the number of monomials the run writes down.*

Proof. The algorithm forms S -polynomials and reduces them. An S -polynomial of f and g is $m_f f - m_g g$ for monomials m_f, m_g , which is a composition of multiplications by variables followed by one linear combination; a reduction step subtracts a monomial multiple of a basis element, likewise. Every intermediate polynomial produced is therefore derivable by the two rules of Definition 2.1 from the generators, and 1 is among them exactly when the output basis is $\{1\}$. $\square$

Theorem 2.3 (Lower bounds). *Let Γ_n be the arithmetized pigeonhole principle PHP_n^{n-1} . Every F_{PC} refutation of Γ_n has degree $\Omega(n)$, and there is $c > 0$ such that every F_{PC} refutation of Γ_n has size at least 2^{cn} . The corresponding statement holds almost surely for random 3-CNF formulas at constant density above the satisfiability threshold.*

Attribution. The degree bound is Razborov’s [6]; the size bound follows by the size–degree tradeoff of Impagliazzo, Pudlák and Sgall [5], whose paper is titled for this consequence. The random-formula statement is Ben-Sasson and Impagliazzo [1]. The proof system was introduced, and its identification with the Gröbner basis algorithm made, by Clegg, Edmonds and Impagliazzo [2]. $\square$

Corollary 2.4. *Algorithm 1.3 does not run in time polynomial in $|\varphi|$, and no variant of it that decides unsatisfiability by exhibiting 1 in the ideal does. This is unconditional: it assumes no complexity-theoretic hypothesis, and in particular does not depend on $P \neq NP$.*

Proof. On input Γ_n the algorithm’s run is a F_{PC} refutation by Proposition 2.2, and its running time is at least the number of monomials it writes. By Theorem 2.3 that number is $2^{\Omega(n)}$, while $|\Gamma_n|$ is polynomial in n . $\square$

Remark 2.5 (The claim of Version 1 is withdrawn). Corollary 2.4 settles the question the earlier version left open, in the negative. The route from Gröbner bases to $P = NP$ is not merely unproved; it is blocked by a theorem proved in 1999. It should be added that this is a good outcome for the method as mathematics: a technique with an exponential lower bound is a technique whose behaviour is understood.

3 What survives

The lower bounds are worst-case statements, and the algebraic method retains three properties that clause-based methods do not have. These, and not the complexity claim, are the reason to be interested in it.

Theorem 3.1 (Simulation). *F_{PC} polynomially simulates resolution over every field, and the degree of the simulating refutation equals the width of the resolution refutation simulated. Hence nothing is given up by working algebraically.*

Attribution and sketch. Translate a clause $\bigvee_{i \in P} p_i \vee \bigvee_{j \in N} \neg p_j$ to the monomial $\prod_{i \in P} (1 - x_i) \prod_{j \in N} x_j$, which vanishes exactly on the assignments satisfying the clause. A resolution step becomes a bounded number of applications of the two rules. Due to Clegg, Edmonds and Impagliazzo [2]. $\square$

Theorem 3.2 (Strict gain on parity structure). *Let T_n be the Tseitin formula of a bounded-degree expander graph on n vertices with odd total charge. Then T_n has F_{PC} refutations of polynomial size over $\mathbf{F}_2$, while every resolution refutation of T_n has size $2^{\Omega(n)}$.*

Attribution and sketch. Over $\mathbf{F}_2$ the parity constraint at a vertex v is the linear polynomial $\sum_{e \ni v} x_e + \sigma(v)$. Summing these over all vertices makes each edge variable occur twice and cancel, leaving $\sum_v \sigma(v) = 1$: a refutation using one linear combination per vertex and no multiplications whatever. The exponential resolution lower bound is Urquhart's [8]. $\square$

Corollary 3.3. $F_{res} \preceq_p F_{PC}$ but $F_{PC} \not\preceq_p F_{res}$: the algebraic system is strictly stronger than resolution.

Theorem 3.4 (Bounded degree is tractable). *For fixed d there is an algorithm running in time $n^{O(d)}$ which finds a F_{PC} refutation of degree at most d whenever one exists.*

Sketch. Multilinear polynomials of degree $\leq d$ in n variables span a space of dimension $\sum_{j \leq d} \binom{n}{j} = n^{O(d)}$. Closing the input under multiplication by variables and taking linear spans, discarding anything of degree above d , is Gaussian elimination on a matrix of that side length; 1 is derivable in degree d iff it lands in the span [2]. $\square$

Remark 3.5 (The question that replaces the original one). Theorem 3.4 localizes the difficulty. The algebraic setting is not itself the obstruction; *degree* is. Where a family of formulas admits refutations of bounded degree, those refutations are found in polynomial time for each fixed bound, and Theorem 3.2 exhibits a natural family admitting refutations of degree 1. The productive question is therefore:

Which families of propositional formulas admit low-degree Polynomial Calculus refutations?

This is answerable, is being answered, and has the property the original question lacked — that partial progress on it is progress. By contrast, Theorem 2.3 tells us in advance that the answer is not “all of them.”

Remark 3.6 (On the Boolean-ring presentation). Carrying the field equations as explicit generators, as in Definition 1.1, is one of several presentations. Boolean Gröbner basis theory works directly in the quotient ring, and admits coefficient rings that are themselves Boolean rather than fields [7]. This is often substantially better in practice and is the right setting for implementation. It does not affect any statement above: the lower bounds of Theorem 2.3 are proved for the proof system, not for a particular data structure.

4 Summary

Claim of Version 1	Status
The arithmetization τ is correct	Correct; Proposition 1.2.
φ unsatisfiable iff 1 reduces to 0 mod G	Correct; Proposition 1.2.
FGLM computes G in polynomial time	False twice over: FGLM converts an existing basis, and is polynomial in the quotient dimension, which is exponential in n .
Hence $P = NP$	Withdrawn. Corollary 2.4 shows the method is exponential on the pigeonhole family, unconditionally.
Retained	
Algebraic methods subsume clause methods	Theorem 3.1.
Algebraic methods strictly beat resolution somewhere	Theorem 3.2, on Tseitin formulas.
Low-degree refutations are efficiently findable	Theorem 3.4.

This note is a companion to Section 10 of *Depths of a Simulation and Formalized Self-Awareness*, Merged Edition, where the same material is developed inside the formal-system vocabulary $F = (x, y, z)$ and the Tseitin separation is recorded as a second unconditional instance of the Two-System Size-Speedup Principle.

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